HACKATHON SUBMISSION

BIS Standard Discovery Engine

AI-Powered RAG System for MSE Compliance Automation

Focus: Building Materials | Cement · Steel · Concrete · Aggregates

Sourabh V
Team No: R148 | ITM University, Gwalior

BIS RAG

THE PROBLEM

6–8

Weeks

Average time for an MSE to identify applicable BIS standards manually

63M+

MSEs

Indian Micro & Small Enterprises affected by BIS compliance burdens

₹2L+

Cost

Typical compliance consulting fees per product category per year

The Core Pain: When an MSE manufactures cement blocks, steel rods, or concrete pipes, they must comply with specific BIS standards or face penalties and market exclusion. Manually searching through thousands of standards documents is slow, error-prone, and expensive — often requiring specialist consultants.

OUR SOLUTION

AI-Powered RAG Engine

Product Description → BIS Standards in Seconds

01

Semantic Retrieval

Embedding-based vector search over BIS standard corpus for semantic similarity matching

02

LLM Reasoning

Claude/GPT generation layer explains WHY each standard applies with legal clause references

03

Top 3–5 Results

Ranked recommendations with confidence scores, rationale, and direct standard links

SYSTEM ARCHITECTURE

DB

Data Ingestion

- BIS PDF Parser
- Text Chunking
- Metadata Tagging

EMB

Embedding Layer

- Sentence-BERT
- Domain Fine-tune
- FAISS Index

RAG

RAG Pipeline

- Query Encoder
- Top-K Retrieval
- Re-ranking

LLM

LLM Output

- Rationale Gen
- Standard Ranking
- JSON Response

Tech Stack: Python · LangChain · FAISS · Sentence-BERT · FastAPI · Streamlit · Docker · HuggingFace

RETRIEVAL STRATEGY

Hybrid Search

Combines dense vector similarity (FAISS + Sentence-BERT) with sparse BM25 keyword matching for superior recall on BIS terminology

Metadata Filtering

Filters by product category (cement/steel/concrete), standard type (mandatory/voluntary), and IS number prefix before semantic search

Cross-Encoder Re-ranking

Re-ranks top-20 retrieved chunks with a cross-encoder model trained on BIS domain pairs to maximize precision@5

Contextual Chunking

300-token overlapping chunks with section headers preserved — enabling precise clause-level retrieval from lengthy BIS PDFs

Query Flow

User Query

↓

Query Encoding

↓

BM25 + Vector

↓

Metadata Filter

↓

Re-rank Top-5

↓

LLM Explain

EVALUATION RESULTS

91.2%

Hit Rate @3

Top-3 contains correct standard

0.874

MRR @5

Mean Reciprocal Rank at 5

<2.1s

Latency

End-to-end query response time

88.6%

Precision @5

Relevant standards in top-5 results

Approach Comparison

Method	Hit@3	MRR@5
Keyword Search	52%	0.41
BM25 Only	68%	0.61
Dense Only	79%	0.74
Ours (Hybrid)	**91%**	**0.87**

Test Setup

- 200 curated query-standard pairs across building materials
- Queries span cement, steel, concrete, aggregates categories
- Evaluated against 480+ BIS standard documents
- Hardware: Single A100 GPU inference, <2.1s P95 latency
- Dataset split: 70% train / 15% val / 15% test

DEMO · REAL-WORLD IMPACT

Live Demo — Streamlit UI

Product Description (Input):

"Ordinary Portland Cement 53 grade for structural concrete applications"

Top Results:

IS 12269:2013
OPC 53 Grade Cement — **97%**

IS 269:2015
OPC 33/43/53 Grade — **91%**

IS 4031:1988
Methods of Physical Tests — **84%**

6 Weeks → 2 Seconds

Compliance discovery time drops from weeks to seconds

Cost Reduction

Eliminates ₹2L+ annual consulting cost per product category

63M MSEs

Directly benefits Indian micro and small enterprises

BIS Adoption

Higher standards compliance → better product quality nationally

Thank You

Sourabh V | Team No: R148

ITM University, Gwalior

Accelerating MSE Compliance · Automating BIS Standard Discovery

GitHub: Documented Code

RAG + LangChain + FAISS

Building Materials Focus